

Assignment 3 (ELEC 341 L3_ODESolution)

Problem 1:

Using Laplace transforms, solve the following differential equations assuming zero initial conditions.

a. $\frac{dx}{dt} + 7x = 5 \cos 2t$

Assuming $ICs = 0$.

b. $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 5 \sin 3t$

c. $\frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 25x = 10u(t)$

Assume that the forcing functions are zero prior to $t = 0^-$.

a) $x' + 7x = 5 \cos(2t)$

$sX(s) + 7X(s) = \frac{5s}{s^2+4}$

$X(s) = \frac{5s}{(s+7)(s^2+4)} = \frac{A}{s+7} + \frac{Bs+C}{s^2+4} \Rightarrow -\frac{35}{53} \frac{1}{s+7} + \frac{1}{53} \frac{35s+20}{s^2+4}$

$\cancel{As^2} + 4\cancel{As} + \cancel{Bs^2} + 7Bs + \cancel{Cs} + 7C = 5s$

$s^2 \quad A+B=0 \quad B=-A$

$s^1 \quad 7B+C=5$

$s^0 \quad 4A+7C=0 \quad C=-\frac{4}{7}A$

$-7A - \frac{4}{7}A = 5$

$-53A = 35$

$A = -\frac{35}{53}$

$B = \frac{35}{53}$

$C = -\frac{4}{7} \cdot -\frac{35}{53} = \frac{20}{53}$

$C = \frac{20}{53}$

$-\frac{35}{53} \frac{1}{s+7} + \frac{1}{53} \left(\frac{35s}{s^2+4} + \frac{20}{s^2+4} \right)$

$x(t) = \left[-\frac{35}{53} e^{-7t} + \frac{1}{53} (35 \cos(2t) + 10 \sin(2t)) \right] u(t)$

b). $x'' + 6x' + 8x = 5 \sin(3t)$

$s^2 + 6s + 8 = (s+2)(s+4)$

$s^2 X(s) + 6sX(s) + 8X(s) = \frac{15}{s^2+9}$

$X(s) = \frac{15}{(s^2+6s+8)(s^2+9)} = \frac{15}{(s+2)(s+4)(s^2+9)} = \frac{A}{s+2} + \frac{B}{s+4} + \frac{Cs+D}{s^2+9}$

let $s = -2$

$A = \frac{15}{(-2+4)(4+9)} = \frac{15}{26}$

$A = \frac{15}{26}$

let $s = -4$

$B = \frac{15}{(-4+2)(9+9)} = \frac{15}{-36} = -\frac{5}{12}$

$B = -\frac{3}{10}$

let $s = 0$

$\frac{15}{(2)(4)(9)} = \frac{A}{2} + \frac{B}{4} + \frac{D}{9}$

$\frac{15}{72} - \frac{A}{2} - \frac{B}{4} = \frac{D}{9}$

$\frac{15}{72} - \frac{15}{72} + \frac{3}{40} = \frac{D}{9}$

used calculator.

$D = -\frac{3}{65}$

let $s = 1$

$\frac{15}{(3)(5)(10)} = \frac{A}{3} + \frac{B}{5} + \frac{C+D}{10}$

$\frac{2}{10} = \frac{15}{150} - \frac{15}{78} + \frac{3}{50} + \frac{3}{650}$

$C = -\frac{18}{65}$

$X(s) = \frac{15}{26} \frac{1}{s+2} - \frac{3}{10} \frac{1}{s+4} - \frac{1}{65} \left(\frac{18s+3}{s^2+9} \right) \rightarrow \frac{18s}{s^2+9} + \frac{3}{s^2+9} - 18 \cos(3t) - \sin(3t)$

$x(t) = \left[\frac{15}{26} e^{-2t} - \frac{3}{10} e^{-4t} - \frac{18}{65} \cos(3t) - \frac{1}{65} \sin(3t) \right] u(t)$

$$c) \quad x'' + 8x' + 25x = 10p(t).$$

$$s^2 + 8s + 16 + 9 \\ = (s+4)^2 + 9$$

$$s^2 X(s) + 8sX(s) + 25X(s) = \frac{10}{s}$$

$$X(s) = \frac{10}{s(s^2 + 8s + 25)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 8s + 25}$$

$$\cancel{As^2} + \cancel{8As} + 25A + \cancel{Bs^2} + \cancel{Cs} = 10.$$

$$\begin{array}{lcl} s^2 & A + B & = 0 \quad A = -B \\ s^1 & 8A + C & = 0 \quad C = -8A \\ s^0 & 25A & = 10 \end{array}$$

$$A = \frac{2}{5}$$

$$B = -2/5$$

$$C = -16/5$$

$$X(s) = \frac{2}{s} \frac{1}{s} - \frac{2}{s} \left(\frac{s+8}{(s+4)^2 + 9} \right)$$

$$X(s) = \frac{2}{s} \frac{1}{s} - \frac{2}{s} \left(\frac{s+4 + 4}{(s+4)^2 + 9} \right)$$

$$= \frac{2}{s} \frac{1}{s} - \frac{2}{s} \left(\frac{s+4}{(s+4)^2 + 9} + \frac{4}{(s+4)^2 + 9} \right)$$

$$x(t) = \left[\frac{2}{s} - \frac{2}{s} \left(e^{-4t} \cos(3t) + \frac{4}{3} e^{-4t} \sin(3t) \right) \right] p(t)$$

Problem 2:

For each of the following transfer functions, write the corresponding differential equation:

a. $\frac{X(s)}{F(s)} = \frac{7}{s^2 + 5s + 10}$

b. $\frac{X(s)}{F(s)} = \frac{15}{(s+10)(s+11)}$

c. $\frac{X(s)}{F(s)} = \frac{s+3}{s^3 + 11s^2 + 12s + 18}$

$\Rightarrow$ a) $\frac{X(s)}{F(s)} = \frac{7}{s^2 + 5s + 10}$

$\mathcal{L}^{-1}\{s^2 X(s) + 5s X(s) + 10 X(s) = 7 F(s)\}$
 $x'' + 5x' + 10x = 7f(t)$

$\Rightarrow$ b). $\frac{X(s)}{F(s)} = \frac{15}{(s+10)(s+11)} = \frac{15}{s^2 + 21s + 110}$

$\mathcal{L}^{-1}\{s^2 X(s) + 21s X(s) + 110 X(s) = 15 F(s)\}$
 $x'' + 21x' + 110x = 15f(t)$

$\Rightarrow$ c) $\frac{X(s)}{F(s)} = \frac{s+3}{s^3 + 11s^2 + 12s + 18}$

$\mathcal{L}^{-1}\{s^3 X(s) + 11s^2 X(s) + 12s X(s) + 18 X(s) = s F(s) + 3 F(s)\}$
 $x''' + 11x'' + 12x' + 18x = f'(t) + 3f(t)$